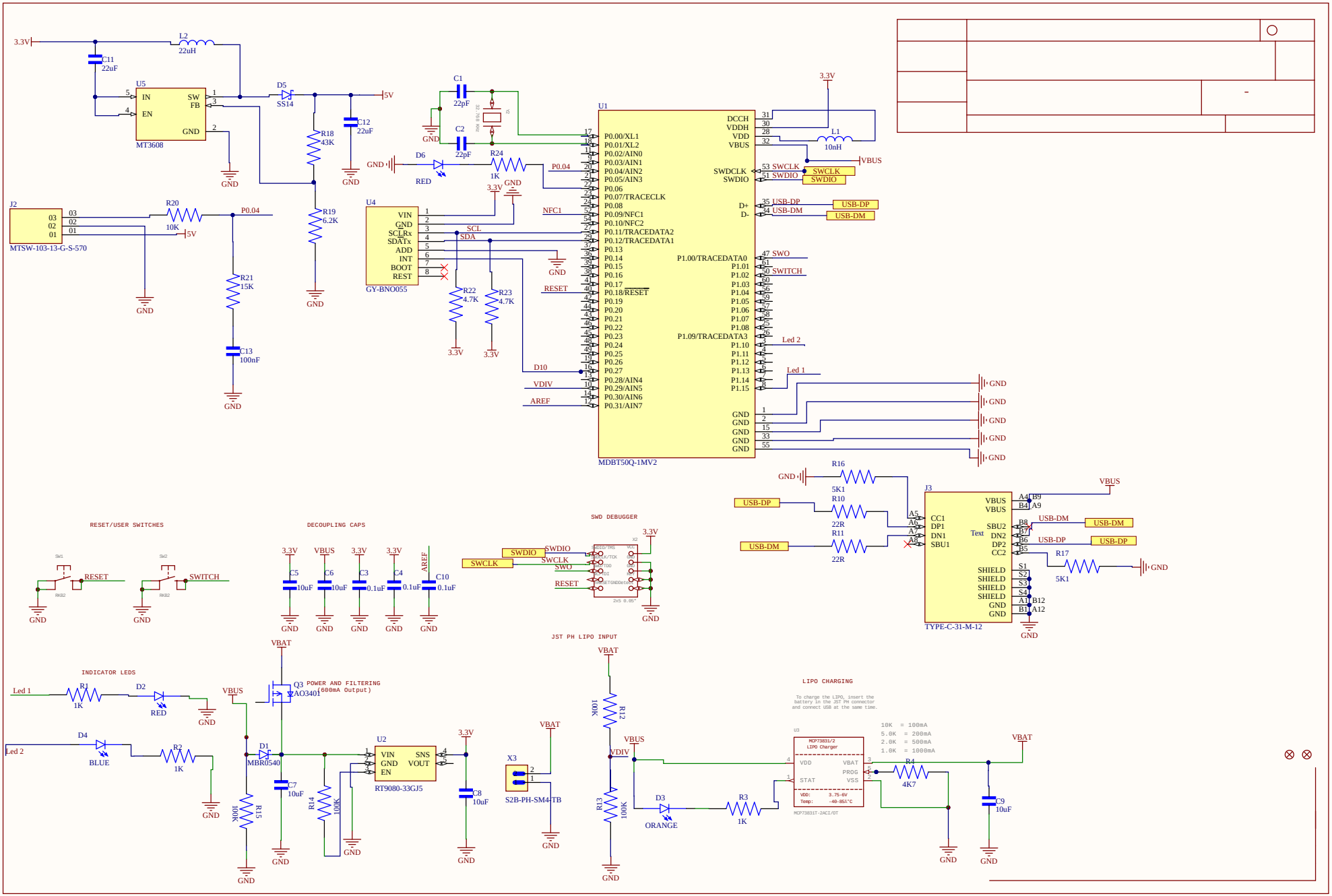
3.3V
L2
22uH
C11
22uF
U5
IN
EN
SW
FB
GND
MT3608
D5
SS14
5V
R18
43K
C12
22uF
R19
6.2K
GND
C1
22pF
C2
22pF
32.768 KHZ
D6
RED
R24
1K
P0.04
NFC1
SCL
SDA
RESET
D10
VDIV
AREF
U1
MDBT50Q-1MV2
DCCH
VDDH
VDD
VBUS
L1
10nH
VBUS
SWDCLK
SWDIO
SWCLK
SWDIO
D+
D-
USB-DP
USB-DM
P1.00/TRACEDATA0
P1.09/TRACEDATA3
SWO
SWITCH
Led 2
Led 1
J2
MTSW-103-13-G-S-570
R20
10K
R21
15K
C13
100nF
U4
GY-BNO055
VIN
GND
SCLRx
SDATx
ADD
INT
BOOT
REST
R22
4.7K
R23
4.7K
R16
5K1
R10
22R
R11
22R
J3
TYPE-C-31-M-12
CC1
DP1
DN1
SBU1
VBUS
SBU2
DN2
DP2
CC2
SHIELD
Text
R17
5K1
RESET/USER SWITCHES
RESET
SWITCH
DECOUPLING CAPS
C5
10uF
C6
10uF
C3
0.1uF
C4
0.1uF
C10
0.1uF
SWD DEBUGGER
SWDIO
SWCLK
SWO
RESET
X2
2x5 0.05"
INDICATOR LEDS
Led 1
R1
1K
D2
RED
Led 2
D4
BLUE
R2
1K
VBAT
Q3
AO3401
POWER AND FILTERING
(600mA Output)
D1
MBR0540
C7
10uF
R15
100K
R14
100K
U2
RT9080-33GJ5
VIN
GND
EN
SNS
VOUT
C8
10uF
JST PH LIPO INPUT
X3
S2B-PH-SM4-TB
R12
100K
R13
100K
VDIV
D3
ORANGE
R3
1K
LIPO CHARGING
To charge the LIPO, insert the battery in the JST PH connector and connect USB at the same time.
10K = 100mA
5.0K = 200mA
2.0K = 500mA
1.0K = 1000mA
U3
MCP73831/2
LIPO Charger
VDD
STAT
VBAT
PROG
VSS
VDD: 3.75-6V
Temp: -40-85ÂºC
MCP73831T-2ACI/OT
R4
4K7
C9
10uF

Comment	Description	Designator	Footprint	LibRef	Quantity
22pF	22 pF Â±5% 50V Ceramic Capacitor C0G, NP0 0603 (1608 Metric)	C1, C2	CAPC1608X90	C0603JPNPO9BN220	2
0.1uF	0.1 Âµf Â±10% 16V Ceramic Capacitor X7R 0603 (1608 Metric)	C3, C4, C10	CAPC1608X90N	C0603C104K4RAC	3
10uF	10 Âµf Â±10% 16V Ceramic Capacitor X5R 0805 (2012 Metric)	C5, C6, C7, C8, C9	CAPC2012X88N	C0805C106K4PACTU	5
22uF	22 µF ±20% 25V Ceramic Capacitor X5R 1206 (3216 Metric)	C11, C12	CAPC3216X259N	12063D226MAT2A	2
100nF	Capacitor MLCC 0603 100nF 50V | AVX 0603SC104K4T2A	C13	CAPC1608X90N	0603SC104K4T2A	1
MBR0540	Diode Schottky 40 V 500mA Surface Mount SOD-123	D1	DIO_MBR0540-TP	MBR0540-TP	1
RED	LED Uni-Color Green 4 Pin Chip 0805 T/R	D2, D6	LEDC2012X70N	597-3305-607F	2
ORANGE	LED Uni-Color Green 4 Pin Chip 0805 T/R	D3	LEDC2012X70N	597-3305-607F	1
BLUE	LED Uni-Color Green 4 Pin Chip 0805 T/R	D4	LEDC2012X70N	597-3305-607F	1
SS14	Diode Schottky 40 V 1A Surface Mount DO-214AC (SMA)	D5	SMA	SS14	1
FIDUCIAL_1MM	Fiducial Alignment Points. Various fiducial points for machine vision alignment.	FID1, FID2	FIDUCIAL_1MM	FIDUCIAL_1MM	2
MTSW-103-13-G-S-570	Connector Header Through Hole 3 position 0.100 (2.54mm)	J2	SAMTEC_MTSW-103-13-G-S-570	MTSW-103-13-G-S-570	1
TYPEC-31-M-12	USB Connectors 24 Receptacle 1 8.94" 7.3mm RoHS	J3	HRO_TYPEC-31-M-12	TYPEC-31-M-12	1
10nH	FIXED IND 10NH 600MA 100 MOHM	L1	INDC2015X152N	LQW15BAS10NG00L	1
22uH	Inductor with Inductance: 22uH Tol. +/-20%, Package: 0805 (2012)	L2	INDC2012X145N	LQM21DH220M70L	1
AO3401	P-Channel 30 V 4A, (Ta) 1.4W (Ta) Surface Mount SOT-23-3	Q3	SOT23-3	AO3401	1
1K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R1, R2, R3, R24	RES_ERJ0G03JR8V	ERJ0G03JR8V	4
4K7	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R4	RES_ERJ0G03JR8V	ERJ0G03JR8V	1
22R	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R10, R11	RES_ERJ0G03JR8V	ERJ0G03JR8V	2
100K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R12, R13, R14, R15	RES_ERJ0G03JR8V	ERJ0G03JR8V	4
5K1	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R16, R17	RES_ERJ0G03JR8V	ERJ0G03JR8V	2
43K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R18	RES_ERJ0G03JR8V	ERJ0G03JR8V	1
6.2K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R19	RES_ERJ0G03JR8V	ERJ0G03JR8V	1
10K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R20	RES_ERJ0G03JR8V	ERJ0G03JR8V	1
15K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R21	RES_ERJ0G03JR8V	ERJ0G03JR8V	1
4.7K	0201 anti-sulfurated resistor 0603 Anti-Sulfurated	R22, R23	RES_ERJ0G03JR8V	ERJ0G03JR8V	2
RKB2	SMT Tact Switches RKB2 https://www.digikey.com/short/mq027pyz5	SW1, SW2	BTN_RKB2_4.6X2.8	SWITCH_TACT_SMT_RKB2	2
MDBT50Q-1MV2	802.15.4, Bluetooth Bluetooth v5.4, ZigbeeÂ®Transceiver Module 2.360GHz – 2.5GHz Integrated, Chip Surface Mount	U1	XCVR_MDBT50Q-1MV2	MDBT50Q-1MV2	1
RT9080-33GJ5	Linear Voltage Regulator ICPositive Fixed 1 Output 600mA TSOT-23-5	U2	SOT94P280X100-5N	RT9080-33GJ5	1
MCP73831T-2ACI/OT	MCP73831/2 LiPO Charger VDD: 3.75 - 6V Temp -40 - 85Â°C Programmable Charge Rate: 15-500mA NOTE: STAT is a tri-state logic output on the MCP73831 and an open-drain output on the MCP73832. SOT23-5 with 4.2V output and tri-state logic on STAT – MCP73831T-2ACI/OT	U3	SOT23-5	MCP73831/2	1
GYBNO055	GYBNO055	U4		GYBNO055	1
MT3608	High Efficiency 1.2MHz2A Step Up Converter	U5	SOT95P280X145-6N	MT3608	1
2x5 0.05"	Serial Wire (SWD) Connector Standard 0.05" 10-pin connector is for use with the Cortex M and Cortex A SWD (serial wire debugger) interface (SW-DP)Connectors 2x05_1.27MM_SMT - Inexpensive surface-mount header with bare pins, compatible with standard 10-pin SWD cables (4UCON: 16846) 2x05_1.27MM_BOX_NOPOSTS - Polarised surface-mount box (4UCON with caps: 19735, without caps: 15117 2x05_1.27MM_BOX_POSTS - Polarised surface-mount box with mounting holes (4UCON: 20910)	X2	2X05_1.27MM_BOX_POSTS	JTAG-CORTEXBOXPOSTS	1
S2B-PH-SM4-TB	Connector Header Surface Mount, Right Angle 2 position 0.079" (2.00mm)	X3	JST_S2B-PH-SM4-TB	S2B-PH-SM4-TB	1
32.768 KHz	Low cost SMT crystals, no capacitors included http://www.ladyada.net/library/eagle	Y2	XTAL3215	XTAL-3.2X1.5	1